



Closed-Loop Feedback Control System for EXOhSPEC Development

Astrophysics Master Project

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Introduction

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- 1.2 Need for Closed Loop Feedback Control
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Literature Review

- 2.1 State of Art: High-Resolution Spectrograph
- 2.2 Importance of Environmental Stability

System Overview

3.1 Introduction to EXOhSPEC

The EXOplanet high-resolution Spectrograph (EXOhSPEC), developed by the University of Hertfordshire, serves as a cutting-edge instrument designed specifically for studying the radial velocities of exoplanets. With a focus on high-resolution spectrometry and precise radial velocity measurement, EXOhSPEC boasts several noteworthy features, including a bifurcated fiber for simultaneous telescope and Thorium Argon input. Central to its functionality is the IDS3010 Displacement Measuring Interferometer, providing picometer-level displacement measurements. Additional sensors, such as the BME680 for pressure and humidity and the PT104 for temperature, have been seamlessly integrated to ensure comprehensive environmental monitoring. The spectrograph achieves a remarkable resolution of $\geq 70,000$ and represents a significant leap in stability optimization through its compact, low-cost, and efficient design.

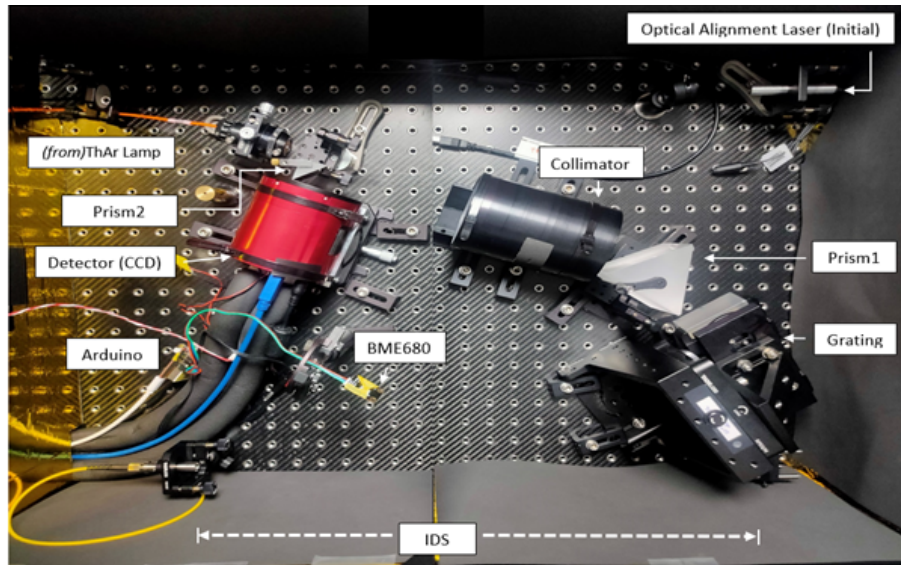


Figure 3.1: *Internal structure of the Modified EXOhSPEC along with IDS and BME680 integrated*

Overview of Spectrograph Operation: The light source utilizes a Thorium Argon lamp, with emitted light channeled through a multimode fiber. The collimator directs incoming light to

Prism 1, which disperses the light spectrum onto the grating for wavelength orders. The grating diffracts light into spectral orders, and the collimator lens focuses the dispersed light. Prism 2 then focuses the spectrum onto the CMOS detector, capturing the spectral characteristics of the incoming light.

3.2 Off-shelf Components Used for Close Loop Feedback Control

Key components include the spectrograph for high-resolution spectral analysis, the IDS 3010 Displacement Sensor for precision interferometric sensing, and environmental monitoring tools such as the BME680 with NODEMCU ESP8266 and the PT-104 Temperature Sensor.

Interferometric Displacement Sensor (IDS3010)

The IDS3010 employs a DFB laser type with a laser power of $400\mu\text{W}$ and a laser wavelength of 1530nm . It serves as a laser displacement measuring interferometer, providing accurate measurements in picometer range for environmental compensation.



Figure 3.2: (a) IDS including Accessories, (b) Sensor head selection and alignment simplified with advanced mounting kits, (c) Optical items: glass target, plane mirror, and retroreflector

BME680

The BME680 is a versatile 4-in-1 digital sensor measuring temperature, humidity, barometric pressure, and gas concentrations like Volatile Organic Compounds (VOC). While gas and temperature are measured, a decision not to use temperature readings from BME680 is explained in section 4.1.3.

Temperature Sensor (PT-104)

The PT-104, functioning as a four-channel temperature-measuring data logger, offers exceptional resolution ($0.001\text{ }^{\circ}\text{C}$) and accuracy ($0.015\text{ }^{\circ}\text{C}$). Utilizing Pico log software for data recording, the information is later processed in Python for calibration purposes.



Figure 3.3: *BME680 module*

Table 3.1: Operating Ranges and Accuracy of Environmental Sensor BME680

Sensor	Operating Range	Accuracy
Temperature	$\pm 1.0\text{ }^{\circ}\text{C}$	-40 to 85 $^{\circ}\text{C}$
Humidity	$\pm 3\%$	0 to 100%
Pressure	$\pm 0.6\text{ hPa}$	300 to 1100 hPa

3.3 Environmental Control Boxes: Box1 and Box2

Box1: Smaller Environmental Control Unit

Box1 serves as an efficient and effective environmental control unit, placing strong emphasis on maintaining stability. It works hand in hand with EXOhSPEC to play a vital role in achieving and maintaining the desired environmental conditions. Enclosed within the box is a spectrograph, securely mounted on a carbon fiber board that ensures thermal stability. The accompanying figure showcases the box, complete with a convenient top lid for effortless configuration adjustments. While it offers precise temperature control, it also allows for airflow, influenced by any temperature changes in the lab setting.

Box2: Modified Protective Box

Box2 is a versatile and ingenious mechanism designed to safeguard the system from external elements. With its advanced temperature regulation and protection features, it guarantees optimal performance and longevity of the equipment. Unlike its smaller counterpart, Box1, Box2 boasts a larger capacity and can accommodate a substantial amount of air. It should be noted, however, that some temperature fluctuations may occur within the enclosure due to the location of the temperature probe at one end. In the future, Box1 will be placed inside Box2 for added protection. Furthermore, an internal fan will be incorporated to promote proper air circulation, resulting in a consistent temperature throughout the enclosure.

3.4 Data Collection Methodology

The block diagram showcases a closed-loop feedback control system, which is affected by external interferences such as temperature, pressure, and humidity. These factors have a direct impact on the optical path length within the system, specifically the distance between the IDS laser head and the retroreflector. Despite a defined expected path length, it is prone to changes due to environmental influences. The closed-loop feedback control system utilizes a structured approach in its data collection methodology to ensure optimum functionality. Its systematic process begins with the initiation phase, where commands are sent through the web interface to activate the



Figure 3.4: *Peltier-104 Temperature Sensor connected with a probe to computer. The temperature variation over time is observed using Pico Log Software*

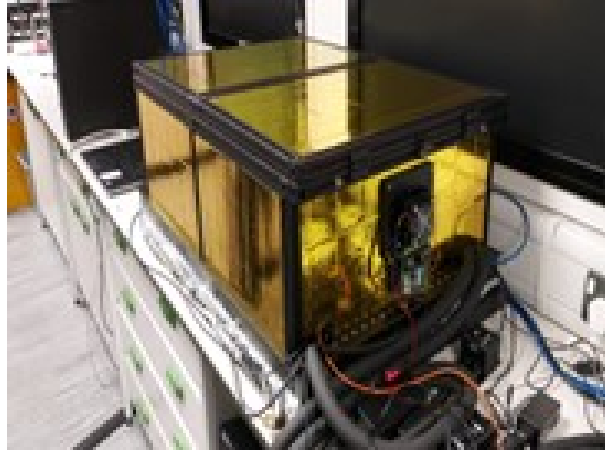


Figure 3.5: *The system's outer box is positioned on a carbon fibre breadboard.*

system. The data acquisition stage utilizes various light sources such as the Thorium Argon Lamp, Sun, or a Star. Accurate measurements of displacement and environmental parameters are carried out using the Interferometer and BME680, respectively, with Python used for data processing. Following acquisition, the sensor data undergoes thorough processing, seamlessly integrating the Interferometer and BME680 data before being transmitted to the Control Unit. Next, the crucial step of spectrograph calibration takes place, involving precise adjustments to the Spectrograph.

The calibration process for EXOhSPEC is tightly linked to its feedback loop, which operates at a rapid pace. Periodic evaluations are necessary to ensure the accuracy and functionality of the system. The outcome of calibration has a significant impact on the next steps taken. In the event of a successful calibration, the system proceeds with data acquisition. On the other hand, if calibration falls short, adjustments are promptly made, and the process is repeated. This rigorous methodology is followed for the desired number of iterations, resulting in a robust and well-calibrated closed-loop feedback system for EXOhSPEC.



Figure 3.6: *Modified protective box designed to shield the system from environmental factors, primarily focusing on temperature control.*

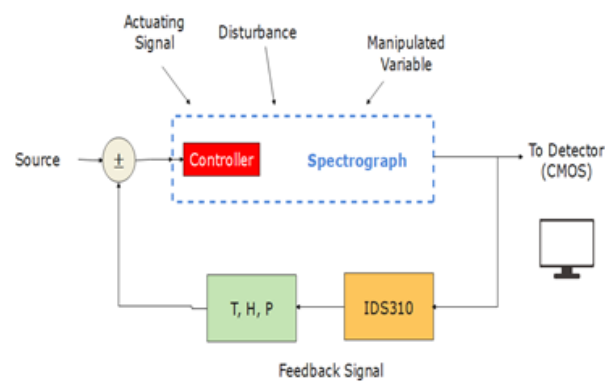


Figure 3.7: *Schematic Block Diagram of the Spectrograph Control Feedback System*

Experimental Data Analysis, Results and Discussion

4.1 Preliminary Testing

The primary objective of the preliminary testing phase was to consolidate data from various proprietary software into a unified platform, specifically using Python. Given that each off-the-shelf component employs its own proprietary software – IDS with WAVE, BME680 with Arduino, and PT106 with Picolog – our initial project phase involved developing Python code to amalgamate all data into one centralized platform. Currently, we retrieve data from Arduino and WAVE software into Jupyter Notebook, saving it as a CSV file for subsequent processing and calibration. Looking forward, our plan is to transition to real-time data collection, ensuring both efficient sampling and consolidated storage within Python. This evolution is aimed at mitigating potential time and space errors, a challenge that will be addressed through the implementation of Python’s multiprocessing and multi-threading modules.

Tuning Spectrograph

In this phase, the focus was on fine-tuning the optical components of the spectrograph to ensure optimal performance. The captured spectral data, shown in the image on the left, reveals bright lines indicating Argon emission lines in the near-infrared spectrum. The exposure time for this spectrum was set at 1 minute, enabling a detailed and comprehensive analysis of the lamp’s emission characteristics. It is crucial to note that variations in temperature, pressure, and humidity can significantly impact the spectrum, particularly when using a high-resolution spectrograph. Calibration becomes imperative to obtain accurate and reliable spectra under changing environmental conditions.

IDS Testing: Positional Displacement Analysis

The IDS3010, a pivotal component, plays a significant role in the project. Retrieving positional displacement data is essential for accurate measurements. The python function `ids.displacement()` is employed to obtain the positional displacement values. Utilizing the fundamental equation for linear displacement laser interferometers,

$$L = \lambda \times \frac{N}{2} \tag{4.1}$$

where N is the number of fringes within the interferometer detectors - initial testing involved plotting raw positional data over time. To enhance the data quality, a moving average, acting

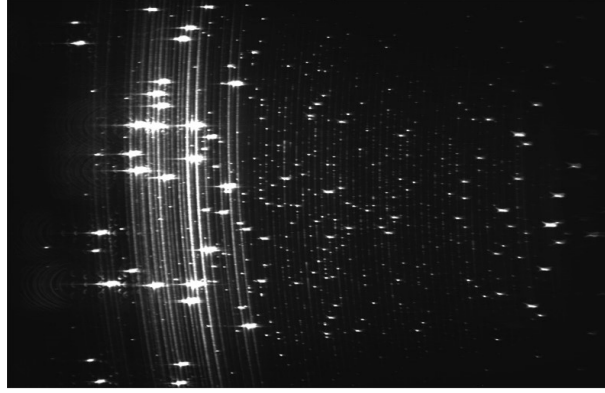
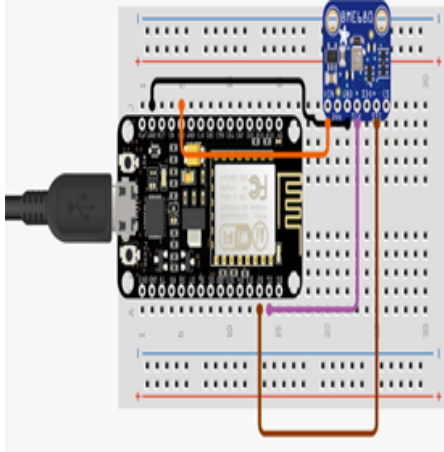


Figure 4.1: *First Calibration and Tuning of ExohSpec result shows Thorium Argon spectrum*

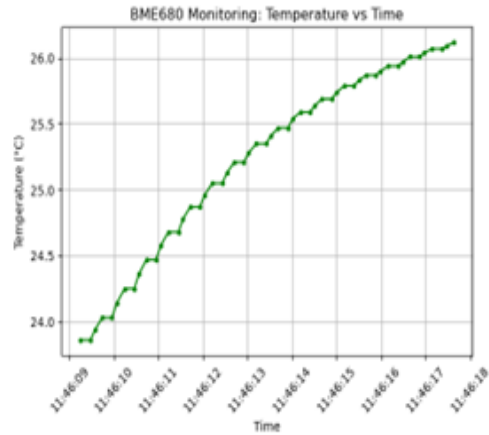
as a low-pass filter, was applied to eliminate high-frequency components, resulting in extracted data points for subsequent calibration.

BME680 Precision Analysis

The subsequent phase of testing focused on the BME680 sensor, revealing an intriguing finding: the temperature readings exhibited fluctuations over time. Further exploration uncovered that the gas element within the BME680 chip operates as a resistor element. This phenomenon can be explained by the fact that the resistivity of conductors increases with rising temperatures. As atoms vibrate more rapidly and over larger distances at higher temperatures, electrons moving through a metal experience more collisions, effectively raising the resistivity. Expressed mathematically, this relationship is captured by the equation $\rho = \rho_0(1 + \alpha \Delta T)$, where ρ_0 represents the original resistivity, α is the temperature coefficient of resistivity, and ΔT is the temperature change. In simpler terms, the resistance of the gas element changes proportionally with temperature fluctuations.



(a)



(b)

Figure 4.2: (a) *Circuit Design of Nodemcu ESP8266 along with BME680 sensor*, (b) *Temperature (°C) vs Time*.

Consequently, the decision was made not to rely on the temperature readings obtained from the BME680 sensor. The reason is that these readings constitute a combination of internal and environmental temperatures from the surrounding environment. Instead, the focus will be on utilizing pressure and humidity readings, which remain reliable. Additionally, it's noteworthy that the accuracy of the BME680 temperature readings is within a range of ± 1 degree Celsius.

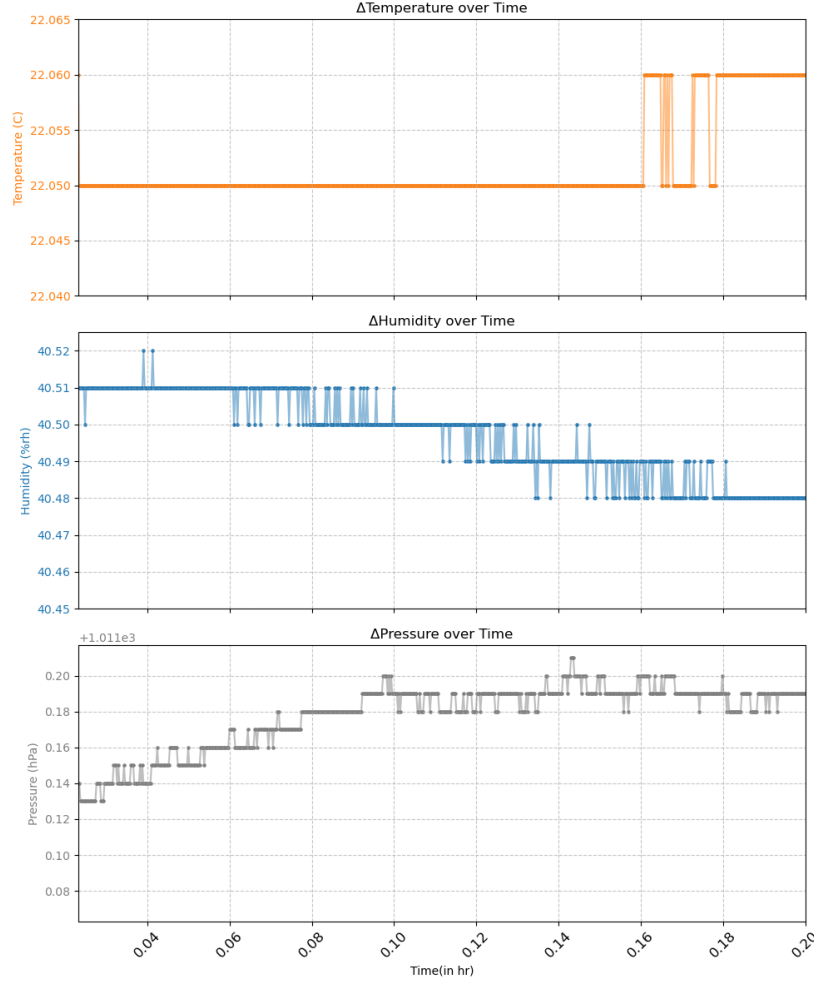


Figure 4.3: *Temperature, Pressure and Humidity variation readings over time in hour taken by BME680.*

To address the temperature accuracy concerns, the project adopted a Peltier-based temperature 4-channel module known as PT104. This module provides significantly more accurate temperature readings compared to the BME680. During the initial testing phase in Box2, a plot was generated, illustrating temperature changes. The experiment involved three probes strategically placed inside the box at different locations, and an additional probe outside the box for reference. Throughout the two-day experiment, the box remained active while the temperature control was turned off. The results revealed crucial insights. Firstly, the temperature within the entire box exhibited non-constant behavior. Secondly, a noticeable change of 2 degrees Celsius from the room temperature was observed. This change was attributed to the internal system operations, leading to the dissipation of heat within the box and, consequently, a rise in temperature. These findings underscore the significance of monitoring and understanding the thermal

dynamics within the experimental setup for accurate environmental control.

PT104

Addressing the temperature variability issue, a Peltier-based temperature 4-channel module (PT104) was introduced. Initial testing in Box2 illustrated temperature changes at different locations inside the box and one external probe. The results highlighted a 2-degree Celsius temperature increase inside the box, attributed to the internal system’s heat dissipation during a two-day activation without temperature control.

4.2 Slip Experiment: Testing Robustness

The conducted experiment aimed to simulate slip conditions, providing insight into the system’s robustness and its response under such scenarios. The collected data will undergo thorough examination, and implications for the performance of EXOhSPEC will be discussed. The experimental setup involved manually opening the box after a specific duration, introducing a slip, and later removing it to assess whether the system would revert to the initial optical path.

Self-Interferometry

Observing the plotted data below, it is evident that the introduction of the slip resulted in a noticeable change in the optical path. However, intriguingly, the optical path did not fully return to its initial position. To comprehend this behavior, it is essential to delve into the concept of self-mixing interferometry employed in the IDS3010. Self-mixing interferometry is a laser-based technique integral to the IDS operation. This method involves the laser not only emitting light but also functioning as a sensor. When the emitted laser light reflects off a surface, a portion of it is scattered back to the laser cavity. This returning light interferes with the light generated by the laser, creating an interference pattern. The interference pattern encodes information about the target’s distance or displacement. In the context of the IDS3010, the initial startup sequence is crucial. At the device’s commencement, specific commands are executed to perform optical alignment. This alignment process ensures that the laser beam is correctly directed and focused on the target. Subsequently, the IDS can provide accurate positional displacement data based on the interference pattern created during its operation. The observed deviation in the optical path following the introduction of the slip suggests that the initial optical alignment, crucial for accurate measurements, might have been perturbed. This deviation highlights the sensitivity of the IDS to changes in the optical path and underscores the significance of precise optical alignment during the device’s startup for reliable positional displacement data.

Slip Width Variation

To introduce additional complexity, two slips of different sizes were employed, yielding specific outcomes. The observed discrepancies in the optical path raise questions about the system’s behavior under varying slip conditions.

Possible Explanation:

- **Hysteresis:** The sensor may exhibit hysteresis, wherein its output depends not only on the current input but also on its previous input history. This behavior may cause a delay or non-reversibility in the response after a change in the optical path.
- **Environmental Effects:** Changes in temperature, humidity, or other environmental factors during the experiment could influence the sensor’s behavior, contributing to variations in the optical path.

- **Mechanical Effects:** The introduction and removal of the glass slip may have induced mechanical changes or stresses in the experimental setup, impacting the sensor’s alignment or calibration.

Potential Solutions:

- **Further Experiment with Long Exposures:** Conduct experiments with extended exposure times to observe the sensor’s response over prolonged periods, potentially capturing hysteresis effects.
- **Recalibrate the Sensor:** After introducing and removing the glass slip, recalibrate the sensor to account for any changes in alignment or calibration.
- **Investigate Potential Hysteresis:** Systematically investigate and analyze the sensor’s response to detect and understand any hysteresis effects.
- **Monitor and Control Environmental Conditions:** Implement rigorous monitoring and control of environmental conditions during experiments to minimize external influences on the sensor.
- **Check for Mechanical Changes:** Examine the experimental setup for any mechanical changes and ensure stability to eliminate potential sources of variation in the optical path.

4.3 Prolonged Operation Assessment

The assessment aimed to gain insights into the impact of prolonged operation on the system, specifically focusing on understanding the refraction of air. The goal was to implement calibration techniques that could effectively compensate for air refraction effects over an extended duration.

The Refraction of Air

The refractive index is a crucial consideration in optical design and metrology. Extensive studies have been conducted on the refractive index of air, recognizing its importance in various applications. In a vacuum, the refractive index of air is standardized to 1. However, under real-world conditions defined by a pressure of 1013.25 mbar, a temperature of 20°C, and a relative humidity of 50%, the refractive index may vary. Understanding these variations is essential in our project, as we operate with a geometric optical path length. Environmental factors can induce nano to picometer-level changes in the geometric path length. Consequently, the output from the IDS represents an uncompensated path length, necessitating the development of calibration techniques to rectify these variations.

Calibration Techniques: Compensation of Air Refraction

To address the effects of air refraction and ensure precise optical path measurements, calibration techniques were employed. In our approach, we utilized NIST’s Metrology Toolbox empirical equation, a modified form of Edlen and Ciddor equations. This equation is commonly employed by various linear displacement laser interferometer environmental compensation units. The NIST equation proves highly useful for estimating environmental compensation errors across a wide range of temperatures, pressures, and humidities. Additionally, it facilitates the application of air refraction compensation to uncompensated laser readings, thereby enhancing the accuracy and reliability of our optical path measurements.

By modified empirical Edlen equation,

$$\eta = 1 + 7.86 \times 10^{-4} \frac{P}{273 + T} - 1.5 \times 10^{-11} RH(T^2 + 160) \quad (4.2)$$

where, T (0 to 35°C) is temperature, P (50 kPa to 120 kPa) is pressure, H (0 to 100%) is humidity, and CO_2 concentration (300 - 600 μ mol/mol) NB: We are not considering gas concentration currently

Now solving for η_{sa} ;

$$\eta_{sa} = \eta_{\text{standard air}} = 1 + 7.86 \times 10^{-4} \frac{P}{272 + T} - 1.5 \times 10^{-11} RH(T^2 + 160) \quad (4.3)$$

$$\begin{aligned} &= 1 + 7.86 \times 10^{-4} \frac{101.325}{273 + 20} - 1.5 \times 10^{-11} \times 50(20^2 + 160) \\ &= 1.00027139838 \end{aligned} \quad (4.4)$$

$\therefore$ Will solve for $L_{\text{compensated}}$ as $L_{\text{compensated}}$ is basically the geometric optical path length;

$$L_{\text{compensated}} = L_{\text{uncompensated}} \times \frac{\eta_{\text{standard air}}}{\eta_{\text{air}}} \quad (4.5)$$

$$L_{\text{compensated}} = L_{\text{uncompensated}} \times \frac{1.0002713938}{\eta_{\text{air}}}$$

4.4 Experiment: Lack of synchronisation

Need for Synchronisation

Improvised Data Collection Techniques

4.5 Experiment: Implementation of Off-shelf components in Box2

System Performance

Temperature Characteristics

4.6 Testing with Slow Model PID Values

PID Calibration

Conclusion

This section will summarize the key findings from the experimental analyses, draw conclusions regarding the performance of the closed-loop feedback control system, and propose avenues for future research and enhancements to EXOhSPEC.

5.1 Summary of Findings

5.2 Implications for Astrophysical Observations

5.3 Limitations and Areas for Improvement

5.4 Future Developments and Enhancements

Appendix

6.1 Initial Project Plan

A comprehensive overview of the project's initial plan, including objectives, methodologies, and expected outcomes.

6.2 Project Timeline

A detailed timeline outlining the various project milestones and activities undertaken over the course of its development.

6.3 Code: Implementation Details and Source Code

Selected portions of the codebase used in the development of the closed-loop feedback control system for EXOhSPEC, providing transparency and facilitating potential future modifications.